

Object cache file (object_cache_file): To speed up operations, the *nagios* service caches the read object definitions and configures them in a cache file, which is then read by the CGI. This also prevents inconsistencies, such as when an object file is being modified, and is saved before all changes are completed.

Status file (status_file): This file is where the status of all monitored hosts and services is stored by Nagios, to be processed by the CGI scripts.

Resource file (resource_file): This parameter too can be specified multiple times. Resource files contain macros that are expanded by Nagios when executing a command found in the commands file. We can look at this in detail below. The CGIs do not read these files, and they can contain sensitive information such as user names and passwords. Therefore, restrictive permissions such as 600 (only the owner can read/write) should be placed on these files. As you can see, the Nagios RPMs install these files in a separate directory, */etc/nagios/private*, which is owned by the root user and readable by the Nagios group:

```
[root@prefect ~]# ls -ld /etc/nagios/
drwxr-xr-x 3 root root 4096 May 21 08:31 /etc/nagios/
[root@prefect ~]# ls -ld /etc/nagios/private/
drwxr-x--- 2 root nagios 4096 May 20 07:35 /etc/nagios/private/
```

The object and resource definition files

Objects are entities that need to be monitored, or are used for monitoring. Some examples are commands, hosts, groups, services and contacts. Let us explore a host object and a command object in this article. In later articles, we can explore other objects in greater detail.

Host object definitions are used to define a particular host that is being monitored; the mandatory directives are:

- **host_name:** a short name for the host. Multiple services can be monitored on a single host. Normally, the FQDN is used.
- **alias:** a longer description.
- **address:** the IP address of the host being monitored.
- **max_check_attempts:** the number of attempts to check the host, if a non-OK state is returned.
- **check_period:** the period name (which is also defined), during which checks should be made.
- **contact_groups:** the contact groups (people to be contacted) in case of problems (or recoveries) with this host.
- **notification_interval:** the time interval (by default, in minutes) after which notifications will be sent, in case the host is still down.
- **notification_period:** the time period in which notifications should be sent. In case the host is down in a time period

that is not in this period, no notifications will be sent

- **notification_options:** This directive can have the following values:
 - **d:** send notifications when the host is down
 - **u:** send notifications if the host is unreachable
 - **r:** send notifications on recoveries
 - **f:** when the host starts and stops flapping (flapping is usually used to determine whether a service/host is stable. Flapping occurs when a service/host changes states too frequently.)
 - **n:** no notifications will be sent

A more efficient way to use host definitions is to define templates and use them. A snippet from the file */etc/nagios/localhost.cfg* that defines a template, and then uses it for a host object definition, is shown below:

```
define host{
    use linux-server          ; Name of host template to
                                use
                                ; This host
                                definition will inherit all variables that are defined
                                ; in (or inherited
                                by) the linux-server host template definition.
    host_name localhost
    alias localhost
    address 127.0.0.1
}
```

The *use* statement above specifies that this host definition uses a template called *linux-server*. It is defined in the same file, as follows:

```
define host{
    name linux-server        ; The name of this host template
    use generic-host         ; This template inherits other
                                values from the generic-host template
    check_period 24x7         ; By default, Linux hosts are checked
                                round the clock
    max_check_attempts 10    ; Check each Linux host 10 times
                                (max)
    check_command check-host-alive ; Default command to check
Linux hosts
    notification_period workhours ; Linux admins hate to
                                be woken; only notify in the day
                                ; Note that notification_period overrides the value
                                ; inherited from the generic-host template!
    notification_interval 120 ; Resend notification
                                every 2 hours
    notification_options d,u,r ; Only send
                                notifications for specific host states
    contact_groups admins      ; Notifications get
                                sent to the admins by default
    register 0                 ; DONT
```